

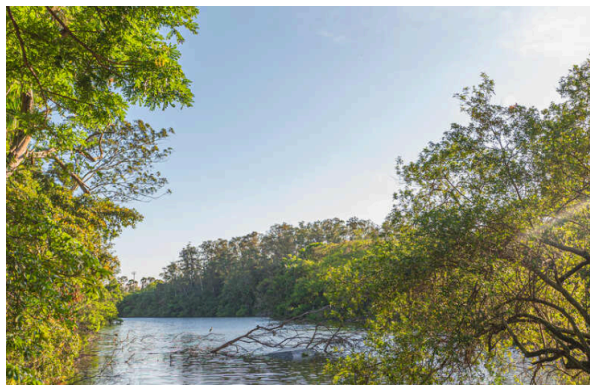
22 Energy in the Ecosystem

Ecosystem Concepts and Energy Flow

- **Ecosystem** function obeys **thermodynamic principles**; energy transformations in ecosystems follow the laws of thermodynamics.
- **Feeding relationships** link organisms into a **food web**, forming a single functional entity.
- The **ecosystem** is the fundamental unit of ecological organization, integrating biological and physical components.
- **Energy** enters ecosystems primarily as **sunlight** and is converted to **chemical energy** by **photosynthesis**.
- **Trophic levels** represent steps in the food chain: **producers**, **herbivores**, **carnivores**.
- Energy is lost at each trophic level due to **respiration** and **inefficiency** of biological processes.
- The **pyramid of energy** illustrates diminishing energy at higher trophic levels:



- **Odum's energy flow model** shows how energy is reduced by respiration and excretion at each trophic level:

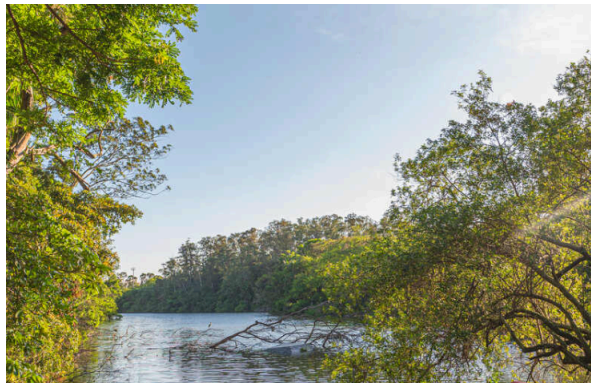


Primary Production

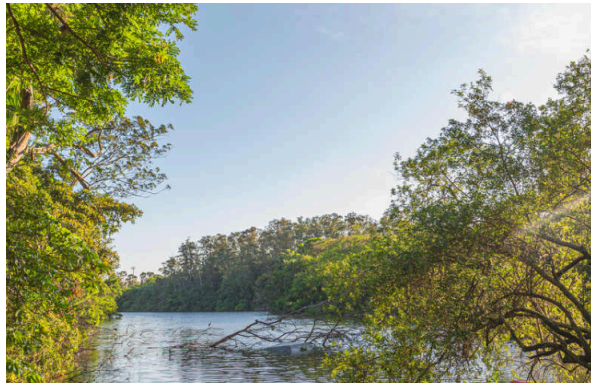
- **Primary production** is the process by which **plants**, **algae**, and some **bacteria** capture light energy and transform it into **carbohydrates** by **photosynthesis**.
- The overall chemical equation for photosynthesis is: $6\text{CO}_2 + 6\text{H}_2\text{O} \longrightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$
- **Gross primary production** is the total energy assimilated by photosynthesis.
- **Net primary production** is the energy accumulated in plant biomass and available to consumers.
- Partitioning of gross primary production:



- Many fluxes are involved in primary production, including **CO₂ uptake**, **O₂ production**, and **biomass accumulation**:



- **CO₂ flux measurements** in light and dark conditions estimate gross primary production:



- In aquatic systems, the **light and dark bottle method** measures production by phytoplankton:

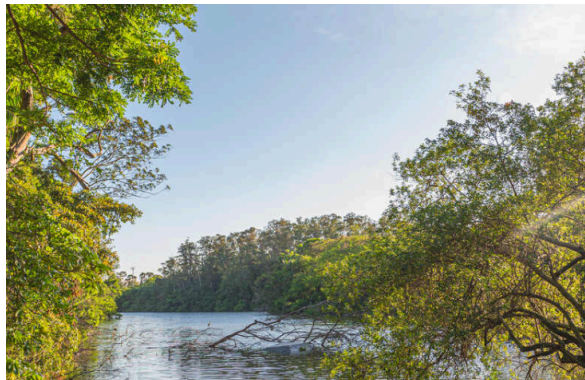


Factors Influencing Primary Production

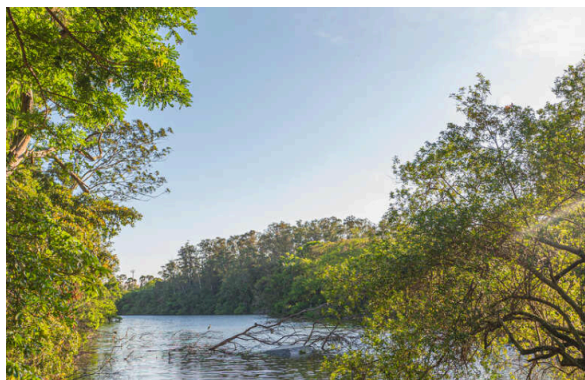
- **Light** and **temperature** affect the rate of photosynthesis; photosynthesis increases with light intensity up to a **saturation point**.
- The **compensation point** is where photosynthesis balances respiration.
- Photosynthetic efficiency is typically 1–2% of incoming sunlight.
- Photosynthesis vs. light intensity:



- **Water** limits primary production in many habitats; **water use efficiency** is the ratio of dry matter produced to water transpired.
- **C₄** and **CAM** photosynthesis increase water use efficiency under hot, dry conditions.
- **Nutrient availability** (especially **nitrogen** and **phosphorus**) limits production; fertilizers stimulate plant growth:

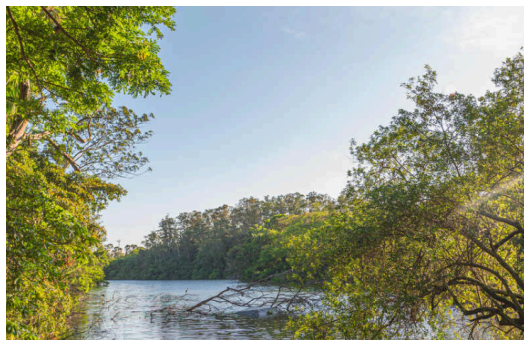


- **Nutrient use efficiency** (NUE) is the ratio of dry matter production to nutrient assimilation; varies across ecosystems:

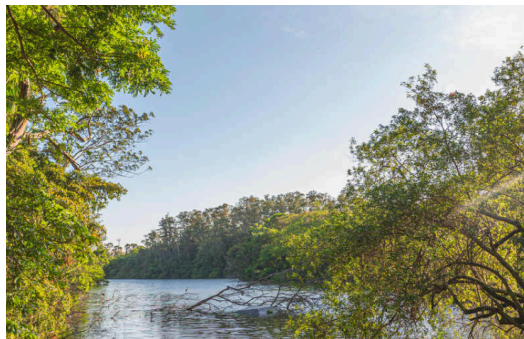


Variation in Primary Production

- **Net primary productivity (NPP)** is highest in the **humid tropics** and varies with **temperature** and **precipitation**.
- Global map and graphs of NPP:



- NPP varies among habitat types:

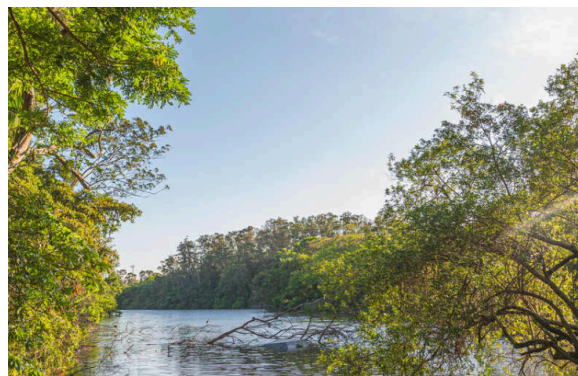


Energy Transfer and Trophic Efficiency

- Only 5–20% of assimilated energy passes between trophic levels; most energy is lost as heat or used for maintenance.
- **Ecological efficiency** or **food chain efficiency** is the percentage of energy transferred from one trophic level to the next.
- **Assimilation efficiency** is the ratio of assimilated energy to ingested energy; varies with food quality.
- Not all food components can be assimilated:



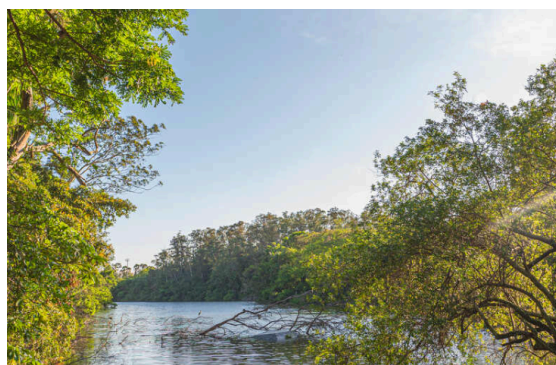
- **Stoichiometry** affects assimilation efficiency; nutrient ratios in food vs. consumer tissues influence waste production.
- Nutrient requirements vary with biology and growth rates:



- **Net production efficiency** is the percentage of assimilated energy used for growth and reproduction.

Detritus Food Chains and Energy Movement

- **Detritus food chains** process dead plant material and indigestible excreta, especially important in terrestrial ecosystems.
- The proportion of NPP entering herbivore vs. detritus food chains varies among ecosystems.
- **Residence time** of energy at each trophic level is the ratio of energy stored to net productivity.
- Biomass accumulation ratios for primary producers:



Ecosystem Energetics and Carbon Balance

- **Ecosystem energetics** summarizes energy movement, trophic structure, and turnover rates.
- **Autochthonous production** is energy produced within the system; **allochthonous inputs** come from outside.
- **Net ecosystem production** is the balance of carbon gain and loss; positive values indicate carbon sinks.
- Cedar Bog Lake energy flow study:

